Calendar

REAL VARIABLES I

Fall 2026

I will put all deadlines on your Blackboard Calendar. Please check it regularly!

Key university dates

For official University dates, see the Fall 2026 academic calendar and the Registrar's final-exam schedule.

Date	Event
Monday, August 17	First day of classes
Friday, August 21	Last day to add this course through the standard add period
Friday, August 28	Last day to drop without a W
Thursday, December 3	Last day of classes and last day to withdraw from all classes

Exams

Date	Event
Through October 27	Oral exam 1 appointments must be completed by this date
October 21, 2026	Take-home written midterm due on Gradescope (120 minutes plus 15
Wednesday, December 9, 2026, 10:15 - 12:15	Final-exam period

Daily lecture log

Wk	Lec	Topics
1	**Riemann integration and its limitations** 2026-08-17	Review: the Riemann integ
	Countability and unbounded functions 2026-08-19	Proposition + proof: conti
	Countability, the extended reals, and sums of non-negative numbers 2026-08-21	Review: fat rationals; sketc
2	**Sums of non-negative numbers** 2026-08-24	Review: continuity of arbit
	Outer measures generated by covering costs 2026-08-26	Motivation: construct a no
	Exercise session: Monotone and dominated convergence for sums 2026-08-28	Exercise session: Monotone
3	**Lebesgue outer measure on $\mathbb{R}^d$** 2026-08-31	Prop+proof: maximality o
	Lebesgue outer measure and the Vitali obstruction 2026-09-02	Prop+proof: translation in
	Exercise session: HW02 2026-09-04	Conclusion of proof: ration
4	**No class: Labor Day** 2026-09-07	No class.
	Measures and their basic inequalities 2026-09-09 TENTATIVE	Defn: measures and measu
	Carathéodory measurable sets, σ-algebras and measures: examples 2026-09-11	Thm+proof(conclusion): C
5	**Dynkin systems and measurable functions** 2026-09-14	Defns: finite, probability, a
	Measurable functions (continued) 2026-09-16	Continuation of the previou
	Exercise session 2026-09-18	Exercise session.
6	**The Lebesgue and Archimedean integrals: basic definitions** 2026-09-21	Defns: superlevel distributi
	Monotone convergence and positive linearity 2026-09-23	Prop+proof: linearity of th
	Homework exercise session 2026-09-25	Homework exercise session.
7	**Properties of the Lebesgue integral: Markov's inequality and null sets** 2026-09-28	Example: an unbounded fu
	Uniform integrability and integrable signed functions 2026-09-30	Defn: uniform integrability
	Exercise session: HW06 2026-10-02	Exercise session: HW06 (th
8	**Dominated convergence and Fatou's lemma** 2026-10-05	Thm (statement): Lebesgu
	Egorov's theorem and Vitali's convergence theorem 2026-10-07	Thm+proof: Egorov's theo
	Exercise session: HW07 2026-10-09	Exercise session: HW07 (re
9	**Riemann and Lebesgue integrals: semicontinuity and envelopes** 2026-10-12	Thm (statement): Lebesgu
	Proof of Lebesgue's criterion 2026-10-14	Defn: infimal convolution.

Tentative. Future lecture topics and their placement may change as the course develops.